

Project Overview Document: Online Book Review Application (BookVerse)

1. Introduction, Objectives, and Scope

Introduction

The BookVerse application is an interactive online platform designed to cater to book enthusiasts. It provides users with the ability to explore new books, read existing reviews, and contribute their own insights. The backend of this application is built using Node.js, Express.js, and MongoDB, while the frontend is developed with React.js, ensuring a robust and responsive user experience.

Objectives

- **Centralized and User-Friendly Platform:** To provide a unified and intuitive platform for searching books by ISBN, title, or author.
- **Community Engagement:** To foster an interactive community where users can read and contribute their opinions and reviews on various books.
- **Secure User Authentication:** To ensure data security through a robust authentication system utilizing JSON Web Tokens (JWT).
- **Developer-Friendly API:** To provide a well-documented API using Swagger UI (OpenAPI Specification) to facilitate seamless integration for both backend and frontend developers.

Scope

The project encompasses the following core functionalities:

1. **User Management:** This includes user registration, secure login, and profile updates.

2. **Book Management:** Functionality for publishing new books to the platform, listing available books with pagination, and advanced searching capabilities.
3. **Review System:** Creation, updating, and deletion of personalized book reviews.
4. **API Documentation:** A fully functional interactive UI for testing and viewing API endpoints.

2. Entities Involved and Their Roles

Based on the system's database architecture, the application revolves around three main entities:

2.1. User

- **Role:** The user is the primary interacting entity with the platform. Users can manage their profiles, contribute new books, and write reviews.
- **Key Attributes:**
 - **Username and Password:** (Securely hashed for security).
 - **Name, Email, and Profile Picture URL.**
- **Permissions:** Only authenticated users can add books and manage (create, edit, delete) their own reviews.

2.2. Book

- **Role:** The core entity of the application. Books are added by users and are available globally for others to search, view, and review.
- **Key Attributes:**
 - **ISBN:** International Standard Book Number, used as a unique identifier for exact searches.
 - **Title:** The title of the book.
 - **Author:** The writer of the book.

2.3. Review

- **Role:** A relationship entity bridging the User and the Book. It represents a specific user's opinion and experience regarding a specific book.
- **Key Attributes:**
 - **Review Text:** The actual content of the user's opinion.
 - **Book Reference:** Links the review to a specific book.
 - **User Reference:** Links the review to its author.
- **Permissions:** Reviews are strictly tied to their authors. A review can only be modified or deleted by the user who originally created it.